

Instructor: Date:

Learner: Regional Branch:

Program: Center:

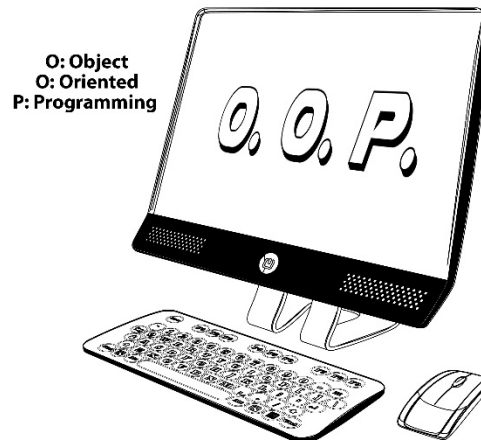
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Let's Start!

Let's learn something about Class-Based programming



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Talk about particular topics related to software programming.

Can Do Statements:

- I can explain the concept of class based programming.
- I can give detailed information about class based programming.
- I can identify important features related to class based programming.

Ask your partner:

- What types of Object-Oriented Programming (OOP) styles do you know?
- Have you used any OOP style? Which one?





Let's Explore!

Nate updates Susan on the CBP lesson



Instructions: Read the text below.

Nate: Hey, Susan! Why weren't you in class today? You missed some important information about Class-based programming.

Susan: Oh really? I thought we were studying that topic next week. What did I miss?

Nate: Well, first, we learned that class-based programming is a style of OOP where the classes of objects are the ones that give inheritance to other objects. This style of programming classifies each object in a class according to its features.

Susan: Oh okay, that's easy to understand. I remember something about encapsulation, was it part of class-based programming?

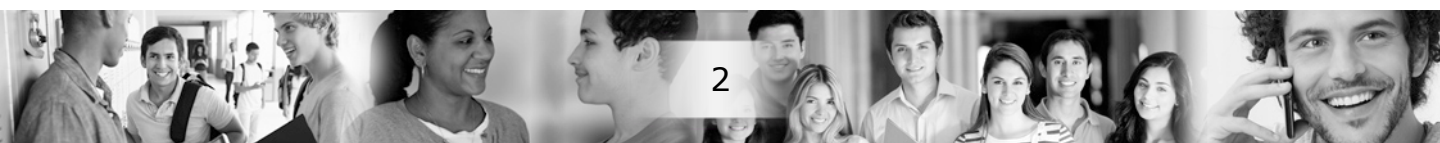
Nate: Yes, it is a feature from class-based OOP. It prevents the users from breaking the invariants of a class. This is important because it focuses on grouping and packaging information instead of any security issues, and this allows you to change aspects of the implementation of an object's class without impacting the users' code.

Susan: Hmm... I need more clarification on each feature. For example, inheritance. Did he explain anything about it?

Nate: Yes, inheritance is done by defining new classes as extensions of an already existing class. The first class is the parent class, and the new one is a child class. If the child class has only one parent class, it is single inheritance, but if it has more, it is multiple inheritance.

Susan: Can you organize classes into a hierarchy?

Nate: Yes, you can organize classes either in "tree", if it is single inheritance, or in "lattice", if it's multiple inheritance.





Let's Practice!

Nate was helping Susan to catch up*.

*To "catch up" means to reach someone's level or standard.

Instructions: Write the words given in the corresponding blanks to correctly complete the sentences.

finished / was saying / were studying / missed

Nate: Hey, Susan! Why weren't you in class today? You _____ really important information about Class-based programming.

Susan: Oh really? I thought we _____ that topic next week. What did I miss?

Nate: Well, first we learned that Class-based programming is a style of OOP where the classes of objects are the ones that give inheritance to other objects. You didn't miss much though! When he _____ this style of programming classifies each object in a class according to its features, the class _____.

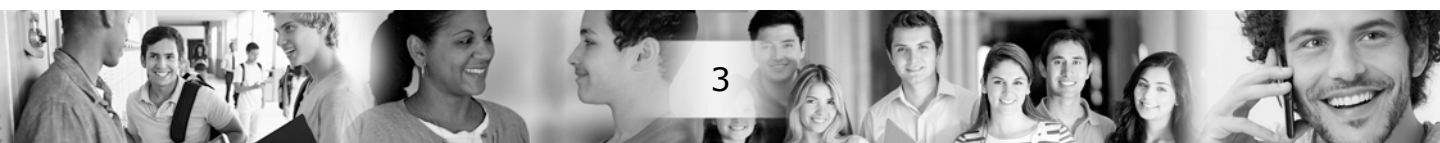


Let's Practice!

Do you want to do a quick review on important features related to class based programming?

Instructions: Underline the correct answer.

1. It might be necessary to provide you with some clarification on (either / each / nothing) feature.
2. (Each / Each concept / Nothing) will be clear if you don't periodically review the concepts you study.
3. If the child class has only one parent class, it is single inheritance, but if it has (either / nothing / each) two or more, you define it as multiple inheritance.
4. You can organize them (nothing / each / either) in a tree structure, if it is single inheritance, or in a lattice diagram, if it's multiple inheritance.





Extension Activity

Talking about classes and objects

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for expanded information on classes and objects in OOP.
- Make a brief presentation to your class in which you explain the importance of classes and objects within the languages that support OOP.
- Provide your classmates and instructor with new and relevant information about these concepts.

Self-Assessment

Yes No

- I look for expanded information on classes and objects in OOP.
- I make a brief presentation about the importance of classes and objects.
- I provide my classmates and instructor with new and relevant information about these concepts.

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Target Words

- CBP** (noun): class-based programming; a style of OOP that divides the objects into classes according to their characteristics
e.g. The Class-based programming is a style of OOP where the classes of objects are the ones that give inheritance to other objects.
- to break** (verb): to interrupt the regularity from a class in CBP
e.g. It prevents the users from breaking the invariants of a class.
- parent class** (noun): main class from CBP that can define new subclasses according to specific characteristics from the main class
e.g. The first class is the parent class.
- style** (noun): set of rules used to write a source code for a computer program
e.g. This style of programming classifies each object in a class according to its features.
- OOP** (noun): object oriented programming; a programming language model organized around objects and not mathematical logic
e.g. The Class-based programming is a style of OOP where the classes of objects are the ones that give inheritance to other objects.



Learning Points

Past Continuous and Simple Past

You use **past continuous** and **simple past** to talk about things that happened in the past and finished in the past, too.

Past Continuous

Use **past continuous** when you want to talk about something that was in progress at a specific time in the past. With this tense, you provide a more extended context to a story or situation.

Follow this structure for affirmative statements:

subject + be in past (**was/were**) + main verb with **ing** + complement

Example:

*They **were programming** when the boss called for a meeting.*

Simple Past

You can use **simple past** to talk about actions that started and finished at specific moments in the past.

This is the structure you use to make affirmative sentences:

subject + **main verb in past** + complement

Example:

*We **worked** for 3 months on that program.*

Examples

- *Why weren't you in class today?*
- *First, we learned that Class-based programming is a style of OOP where the classes of objects are the ones that give inheritance to other objects.*
- *I thought we were studying that topic next week.*



Learning Points

Specific pronouns with verbs in singular: **Either**, **Each**, and **Nothing**

Use **each**, **either** or **nothing** to compare two things and talk about specific objects.

- Use **either** to express alternatives or a choice between two things.

Example:

*I can use **either** prototype based **or** class-based programming for this project.*

- Use **each** to talk about different factors or parts of a whole as individuals.

Example:

***Each** object can have different classes.*

- Use **nothing** as an indefinite pronoun to show the absence of a thing in affirmative sentences.

Example:

*There was **nothing** to do about the programming language they chose.*

Examples

- *This style of programming classifies each object in a class according to its features.*
- *I need more clarification on each feature*
- *You can organize classes either in tree, if it is single inheritance, or in lattice, if it's multiple inheritance.*



Learning Points

Programming and computer science terms

There are certain terms you need to understand from programming and computer science.

Fat Binary: It is a computer executable program that is possible to run in multiple processor types because it is expanded with a native code.

Filename Extension: It is the suffix you give to a computer file. This indicates the characteristics of the file and the programs that can run it.

First Generation Programming Language: This language is a grouping of programming languages you use to program first generation computers. The instructions made in 1GL (First Generation Programming Language) were in binary numbers.

FTP Client: It is a software service that allows the connection between a remote server and a host computer. It permits a dual direction transfer of data and files.

Functional Programming: It is a style used to build the structure and elements of a computer program. It is a declarative programming paradigm; for this reason, you use it with declarations and expressions instead of statements.

